Shared-split baseline appendix

22 September 2026

A Shared-split baseline comparisons

The first two tables report Spearman correlations with LRE faithfulness across 25 equally weighted relations. These runs predate the final-normalization fix merged on 22 September 2026: Gemma2 requires a corrected rerun. Exact executed sources are archived; these results do not validate the merged code.

Matched protocol. After filtering objects whose first answer token does not identify a unique complete object, seed 0 selects eight demonstrations, eight references, and up to 100 test rows. All readouts receive the same test pool within a model; tokenizer-dependent filtering can change membership across models. The neutral prompt is exactly `\n{subject} is known for`. Models use pinned revisions, float32, eager attention, and one fixed layer per model. The relation prompts, model answer tokens, and selected LRE settings are held fixed.

Table 1: **Readout baselines across seven models.** Spearman correlation ρ with LRE faithfulness on 25 shared-split relations per model. All neural readouts use the same newline-prefixed neutral prompt.

Model	Differential Jacobian	Raw Jacobian	Logit lens	Differential logit lens	Attribute rate
OLMo 3 7B	0.683	**0.689**	0.234	0.326	−0.114
GPT-J 6B	**0.573**	0.567	0.531	0.394	0.002
Mistral 7B	**0.606**	0.586	0.402	0.249	−0.304
Gemma 2 9B	0.554	**0.575**	0.554	0.221	−0.292
Qwen3 8B Base	0.729	0.736	**0.743**	0.607	0.052
Qwen3.8 27B	0.600	**0.644**	0.402	0.454	−0.196
Pythia 6.9B[†]	0.479	**0.501**	−0.343	0.217	0.096

Bold marks the largest point estimate within each model, not a significant difference. [†]Pythia's attribute rate excludes 413 test rows with no predictions surviving the attribute filter; the other models have no undefined rates.

Readout definitions. For subject state s_e, reference states s_i, and reference Jacobians J_i, the differential and raw Jacobian readouts decode $\mathrm{mean}_i\, J_i(s_e - s_i)$ and $\mathrm{mean}_i\, J_i s_e$, respectively. Logit lens decodes s_e; differential logit lens decodes $s_e - \mathrm{mean}_i\, s_i$. Same-subject references are excluded, and final normalization is applied once before the vocabulary head. Each relation's score is the negative median $\log_{10}(\mathrm{rank} + 1)$ of the model's answer token.

Attribute rate. Following Geva et al. (2023) and the reconstruction of Gerstner et al. (2026), we measure overlap between filtered top-50 logit-lens predictions and subject attributes retrieved from Wikipedia. The index uses the union of current test subjects; the source stopword, length, and retrieval filters are retained. Relation scores average the percentage overlap. Undefined rates are handled as noted in the table; matched-row sensitivity results remain in the accompanying data.

Raw Jacobian has a larger point estimate than differential Jacobian in five models. All seven paired 95% relation-bootstrap intervals for their difference include zero; these results do not establish a general advantage for either readout. LRE hyperparameter selection remains in sample.

A.1 OLMo corpus-frequency baselines

Table 2 uses the completed scan of OLMo's three published training mixes: 175,877 shards and 3.86 billion document records. The counts are reused, while all aggregates and correlations are recomputed on the current shared-split test rows. They describe the published document multiset, rather than exact packed training sequences or training exposure weights.

Table 2: **Frequency baselines for OLMo 3 7B.** Correlations with LRE faithfulness across the same 25 shared-split relations, using document counts from OLMo's three published training mixes combined.

Frequency measure	Spearman ρ	p-value
Marginal frequencies		
Subject frequency, c_s	0.290	0.160
Object frequency, c_o	0.644	5.08×10^{-4}
Co-occurrence		
Joint frequency, c_{so}	0.578	0.002
Mean of $\log(1 + c_{so})$	0.568	0.003
$\log(1 + \text{mean}(c_{so}))$	0.578	0.002
Conditional frequencies		
Forward conditional, $P(o \mid s)$	0.386	0.056
Reverse conditional, $P(s \mid o)$	−0.328	0.109
Directional difference, $P(o \mid s) - P(s \mid o)$	0.362	0.075
Micro-averaged $P(o \mid s)$	0.615	0.001
Micro-averaged $P(s \mid o)$	0.030	0.887

Counts denote binary document presence. Means use the current test rows with both marginals observed; micro averages are ratios of summed counts. The p-values are two-sided and unadjusted.

Counting and aggregation. A document contributes at most one mention per subject/object string. Matching uses normalized literal names with Unicode word boundaries, without alias resolution or a proximity constraint. With subject, object, and joint counts c_s, c_o, and c_{so}, the conditionals are $P(o \mid s) = c_{so}/c_s$ and $P(s \mid o) = c_{so}/c_o$. Macro features average row-level quantities where both marginals are observed. Micro conditionals divide summed joint counts by summed marginals. The log of the mean joint count has the same Spearman correlation as the mean joint count because the transformation is monotonic.

Interpretation and reproducibility. These are exploratory associations. Relations can share subjects and objects, and the reported p-values are unadjusted. Conditional direction also reflects marginal frequency: when $c_{so} > 0$, $P(o \mid s)/P(s \mid o) = c_o/c_s$. The full measurements, Pearson results, coverage sensitivities, paired intervals, and source hashes remain in `code/results/shared_baselines/`. The tables are generated directly from the verified correlation records.

A.2 Neutral-prompt variations

Table 3 collects the prompt experiments for the differential Jacobian lens. For Qwen3 8B Base, we compare the original prompt with six alternatives and the subsequently measured newline control. The checkpoint, layer 19, reference and test rows, relation-prompt answer tokens, and LRE outcomes are fixed within each protocol. The legacy split contains 28 relations; the shared split contains 25. The lower panel reports the original-to-newline comparison for all seven models on the shared split, using each model's fixed layer.

Table 3: **Neutral-prompt variations.** Spearman ρ between differential-Jacobian scores and retained LRE faithfulness. Panel A compares exact templates for Qwen3 8B Base; panel B isolates the newline-prefix comparison across all seven models on the shared split.

A. Qwen3 8B Base: prompt wording and prefix

Neutral prompt	Shared ($n = 25$)	Legacy ($n = 28$)
`{s} is known for`	0.268	0.060
`{s}`	−0.057	0.058
`{s} is`	0.166	−0.118
`{s} can be described as`	0.372	0.179
`Information about {s}:`	0.733	0.295
`Facts about {s}:`	0.788	0.298
`Entity: {s} is known for`	0.728	0.202
`\n{s} is known for`	0.729	0.216

B. All models: newline prefix, shared split ($n = 25$)

Model	Original	Newline	$\Delta\rho$
OLMo 3 7B	0.624	0.683	+0.058
GPT-J 6B	0.155	0.573	+0.418
Mistral 7B	0.625	0.606	−0.018
Gemma 2 9B	0.487	0.554	+0.066
Qwen3 8B Base	0.268	0.729	+0.461
Qwen3.8 27B	0.596	0.600	+0.003
Pythia 6.9B	−0.045	0.479	+0.524

s denotes the subject; `\n` denotes one leading newline. Panel B compares `{s} is known for` with `\n{s} is known for`. The newline is a position control measured after the seven-template Qwen comparison; no additional newline is implied for the other templates.

Interpretation. Qwen's shared-split correlation rises from 0.268 under the original prompt to 0.729 with a leading newline; `Facts about {s}:` gives 0.788. These are exploratory sensitivity results, not held-out prompt selection. Prefix wording can change both context and subject position. The newline preserves Qwen's original content token IDs, while Mistral also retokenizes the initial content. The legacy and shared protocols are not independent replications, and no layer or LRE hyperparameter is retuned between prompts.

References

Sebastian Gerstner, Hilal AlQuabeh, Kentaro Inui, and Hinrich Schütze. Weakening neurons: An input-output functionality in transformers with outsize influence. *arXiv preprint arXiv:2609.18612*, 2026. URL https://arxiv.org/abs/2609.18612.

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